Hospital Emergency Room Data Analysis

1. Project Overview

This project analyse hospital emergency room data containing 9,216 patient records. The goal is to understand patient demographics, admission patterns, waiting times, satisfaction scores, and department referrals. The project uses Python for data preparation, PostgreSQL for database integration and SQL analysis, and Power BI to create an interactive dashboard for presenting insights and supporting hospital service planning.

2. Dataset Summary

- **Rows:** 9,216
- **Columns:** 12 original columns
- **Key Features:**

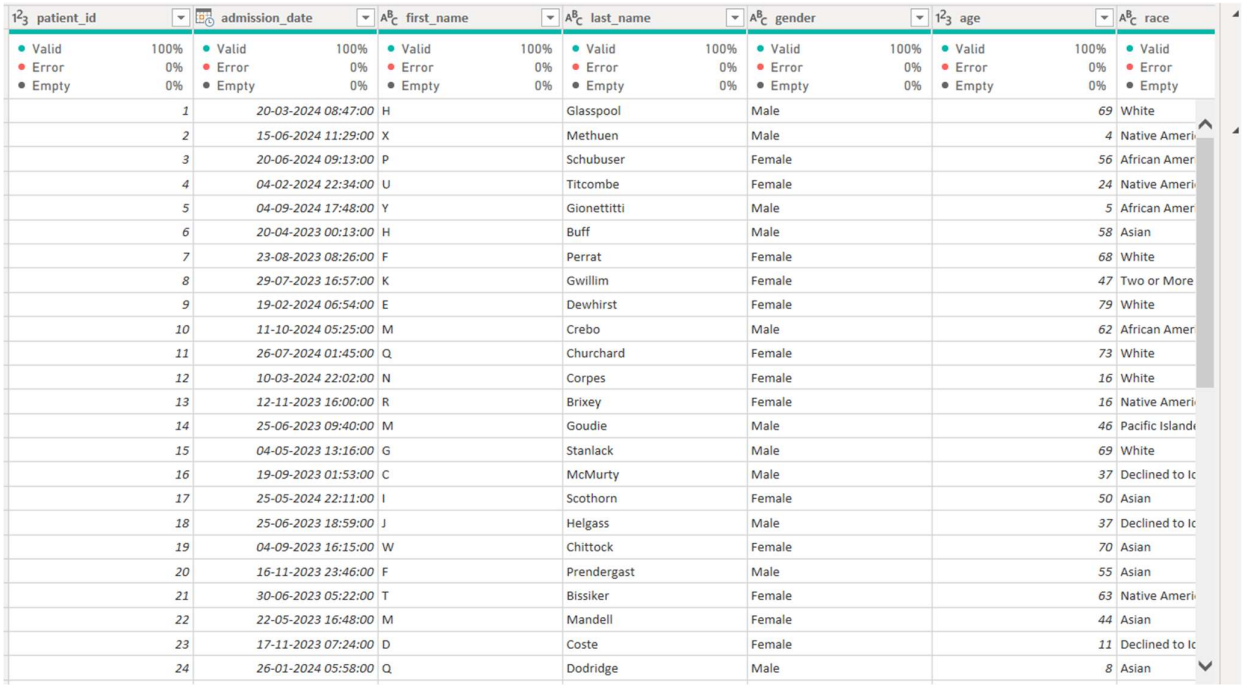
  - Patient demographics: Gender, Age, Race

  - Visit information: Patient ID, Admission Date

  - Hospital details: Department Referral, Admission Flag

  - Patient experience: Satisfaction Score, Waiting Time

- **Missing Data:** Missing values are present in the Department Referral and Patient Satisfaction Score columns.



patient_id	admission_date	first_name	last_name	gender	age	race
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	20-03-2024 08:47:00	H	Glasspool	Male	69	White
2	15-06-2024 11:29:00	X	Methuen	Male	4	Native Ameri
3	20-06-2024 09:13:00	P	Schubuser	Female	56	African Amer
4	04-02-2024 22:34:00	U	Titcombe	Female	24	Native Ameri
5	04-09-2024 17:48:00	Y	Gionettitti	Male	5	African Amer
6	20-04-2023 00:13:00	H	Buff	Male	58	Asian
7	23-08-2023 08:26:00	F	Perrat	Female	68	White
8	29-07-2023 16:57:00	K	Gwillim	Female	47	Two or More
9	19-02-2024 06:54:00	E	Dewhirst	Female	79	White
10	11-10-2024 05:25:00	M	Crebo	Male	62	African Amer
11	26-07-2024 01:45:00	Q	Churchard	Female	73	White
12	10-03-2024 22:02:00	N	Corpes	Female	16	White
13	12-11-2023 16:00:00	R	Brixey	Female	16	Native Ameri
14	25-06-2023 09:40:00	M	Goudie	Male	46	Pacific Island
15	04-05-2023 13:16:00	G	Stanlack	Male	69	White
16	19-09-2023 01:53:00	C	McMurty	Male	37	Declined to Id
17	25-05-2024 22:11:00	I	Scothorn	Female	50	Asian
18	25-06-2023 18:59:00	J	Helgass	Male	37	Declined to Id
19	04-09-2023 16:15:00	W	Chittock	Female	70	Asian
20	16-11-2023 23:46:00	F	Prendergast	Male	55	Asian
21	30-06-2023 05:22:00	T	Bissiker	Female	63	Native Ameri
22	22-05-2023 16:48:00	M	Mandell	Female	44	Asian
23	17-11-2023 07:24:00	D	Coste	Female	11	Declined to Id
24	26-01-2024 05:58:00	Q	Dodridge	Male	8	Asian

3. Exploratory Data Analysis using Python

Data preparation and cleaning were performed to make the dataset suitable for analysis.

- **Data Loading:** Imported the hospital dataset using pandas.
- **Initial Exploration:** Examined the dataset structure, column names, data types, and summary statistics.
- **Data Cleaning:** Removed unnecessary patient name columns and checked for duplicate admission flag information.
- **Missing Data Handling:** Identified missing values in department referral and satisfaction score fields.
- **Column Standardization:** Standardized gender values and improved column naming for readability.
- **Feature Engineering:** Created age groups and prepared date-related fields such as month, quarter, day of the week, and hour for further analysis.
- **Data Type Conversion:** Converted admission dates and waiting time to suitable data types for analysis.
- **Database Integration:** Connected the cleaned dataset to PostgreSQL for structured querying.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df=df.rename(columns={'patient_admission_date':'admission_date'})
df=df.rename(columns={'first_inital':'first_name'})
```

```
df.columns
```

```
Index(['patient_id', 'admission_date', 'first_name', 'last_name', 'gender',
       'age', 'race', 'department_referral', 'admission_flag',
       'satisfaction_score', 'waittime', 'admission_flag.1'],
      dtype='object')
```

```
df.rename(columns={'id': 'patient_id'}, inplace=True)

print(df.columns.tolist())
```

```
['patient_id', 'admission_date', 'first_inital', 'last_name', 'gender', 'age', 'race', 'department_referral', 'admission_flag', 'satisfaction_score', 'wa
ittime', 'admission_flag.1']
```

```
labels=['Young Adult','Adult','Middle-Aged','Senior']
df['age_group']=pd.qcut(df['age'],q=4, labels=labels)
```

```
df[['age','age_group']].head(10)
```

	age	age_group
0	69	Senior
1	4	Young Adult
2	56	Middle-Aged
3	24	Adult

[4]:

	Patient Id	Patient Admission Date	Patient First Inital	Patient Last Name	Patient Gender	Patient Age	Patient Race	Department Referral	Patient Admission Flag	Patient Satisfaction Score	Patient Waittime	Patient Admission Flag.1
0	145-39-5406	20-03-2024 08:47	H	Glasspool	M	69	White	NaN	False	10.0	39	False
1	316-34-3057	15-06-2024 11:29	X	Methuen	M	4	Native American/Alaska Native	NaN	True	NaN	27	True
2	897-46-3852	20-06-2024 09:13	P	Schubuser	F	56	African American	General Practice	True	9.0	55	True
3	358-31-9711	04-02-2024 22:34	U	Titcombe	F	24	Native American/Alaska Native	General Practice	True	8.0	31	True
4	289-26-0537	04-09-2024 17:48	Y	Gionettitti	Male	5	African American	Orthopedics	False	NaN	10	False

[5]:
```
df['Department Referral']=df['Department Referral'].fillna(df['Department Referral'].mode()[0])
```

[6]:
```
df['Patient Satisfaction Score']=df['Patient Satisfaction Score'].fillna(df['Patient Satisfaction Score'].median())
```

[7]:
```
print(df.isnull().sum())
```

```
Patient Id                  0
Patient Admission Date      0
Patient First Inital        0
Patient Last Name           0
Patient Gender              0
Patient Age                 0
```

4. Data Analysis using SQL (Hospital Operations)

SQL analysis can be used to answer important questions about emergency room activity:

1. **Total Patient Visits** – Counted the patient records to understand overall emergency room demand.

```
--Q1. What is the total number of patients by gender?
SELECT
    gender,
    COUNT(*) AS total_patients
FROM "Patient"
GROUP BY gender
ORDER BY total_patients DESC;
```

Data Output | Messages | Notifications

SQL

	gender text	total_patients bigint
1	Male	4729
2	Female	4487

2. **Admission Analysis** – Examined the number and percentage of records marked as admitted.

```sql
--Q5. How many patients were admitted versus not admitted?
SELECT
    admission_flag,
    COUNT(*) AS total_patients,
    ROUND(
        100.0 * COUNT(*) / SUM(COUNT(*)) OVER (),
        2
    ) AS percentage_of_patients
FROM "Patient"
GROUP BY admission_flag
ORDER BY total_patients DESC;
```

Data Output | Messages | Notifications

SQL

	admission_flag boolean	total_patients bigint	percentage_of_patients numeric
1	true	4612	50.04
2	false	4604	49.96

3.Patient Visits by Department – Compared patient counts across department referral categories.

```sql
--Q6. Which 5 department referrals have the highest patient admission rate?
SELECT
    department_referral,
    COUNT(*) AS total_patients,
    SUM(
        CASE
            WHEN admission_flag = 'Yes' THEN 1
            ELSE 0
        END
    ) AS admitted_patients,
    ROUND(
        100.0 * SUM(
            CASE
                WHEN admission_flag = 'Yes' THEN 1
                ELSE 0
            END
        ) / COUNT(*),
        2
    ) AS admission_rate_percentage
FROM "Patient"
WHERE department_referral IS NOT NULL
GROUP BY department_referral
ORDER BY admission_rate_percentage DESC
LIMIT 5;
```

	department_referral text	total_patients bigint	admitted_patients bigint	admission_rate_percentage numeric
1	Renal	86	46	53.49
2	Neurology	193	97	50.26
3	Orthopedics	995	498	50.05
4	General Practice	7240	3623	50.04
5	Gastroenterology	178	89	50.00

4.Waiting Time Analysis – Calculated average waiting time to understand the recorded patient waiting experience.

```sql
--Q2. Which patients had a waiting time greater than the average waiting time?
SELECT
    patient_id,
    age,
    gender,
    waiting_time
FROM "Patient"
WHERE CAST(
    NULLIF(REGEXP_REPLACE(waiting_time::text, '[^0-9.]', '', 'g'), '')
    AS NUMERIC
) > (
    SELECT AVG(
        CAST(
            NULLIF(REGEXP_REPLACE(waiting_time::text, '[^0-9.]', '', 'g'), '')
            AS NUMERIC
        )
    )
    FROM "Patient"
);
```

	patient_id bigint	age bigint	gender text	waiting_time text
1	1	69	Male	39 min
2	3	56	Female	55 min
3	6	58	Male	59 min
4	7	68	Female	43 min
5	9	79	Female	42 min
6	10	62	Male	51 min
7	12	16	Female	39 min
8	13	16	Female	53 min
9	14	46	Male	45 min
10	15	69	Male	49 min
11	16	37	Male	57 min
12	18	37	Male	55 min
13	19	70	Female	50 min
14	20	55	Male	40 min
15	22	44	Female	51 min
16	26	42	Male	51 min
17	27	68	Female	58 min
18	29	58	Female	55 min
19	30	72	Female	37 min

5.Patient Satisfaction Analysis – Examined available satisfaction scores and their average.

```sql
--Q3. Which are the top 5 department referrals with
--the highest average patient satisfaction scores?
SELECT
    department_referral,
    ROUND(AVG(satisfaction_score::NUMERIC), 2)
        AS average_satisfaction
FROM "Patient"
WHERE satisfaction_score IS NOT NULL
GROUP BY department_referral
ORDER BY average_satisfaction DESC
LIMIT 5;
```

SQL

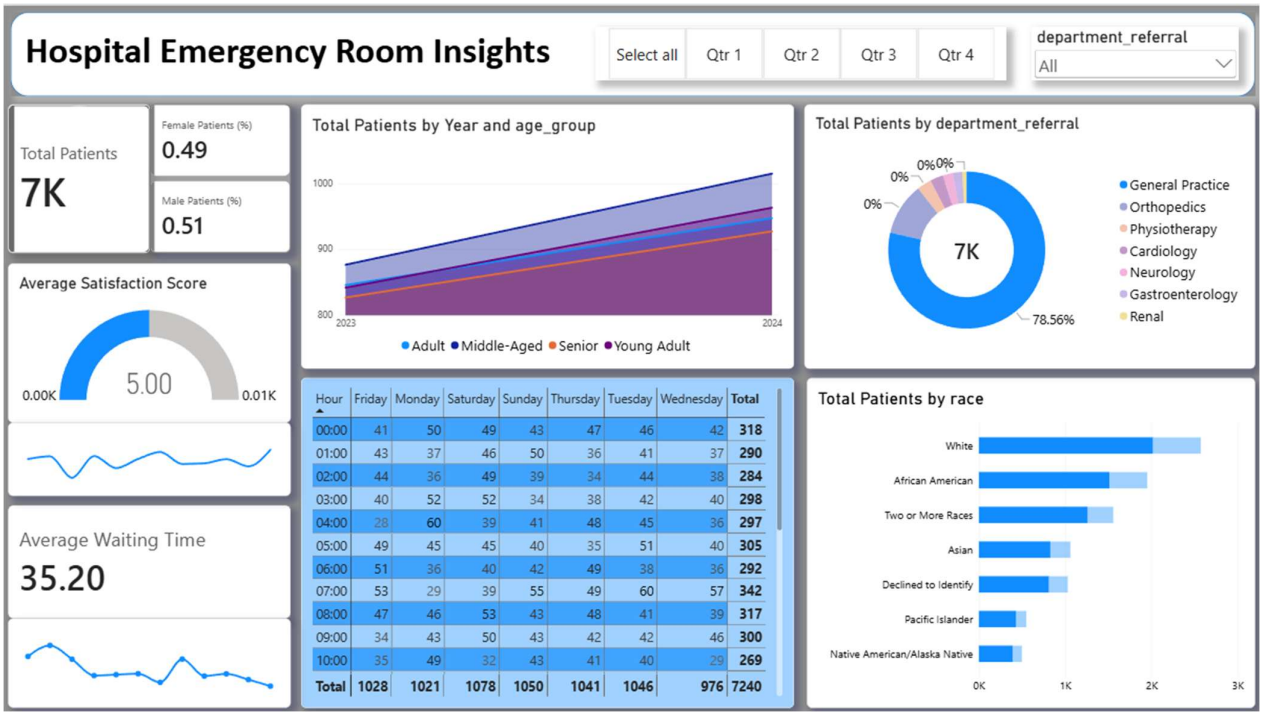
	department_referral text	average_satisfaction numeric
1	Gastroenterology	5.24
2	Neurology	5.08
3	Cardiology	5.04
4	General Practice	5.00
5	Physiotherapy	5.00

5. Dashboard in Power BI

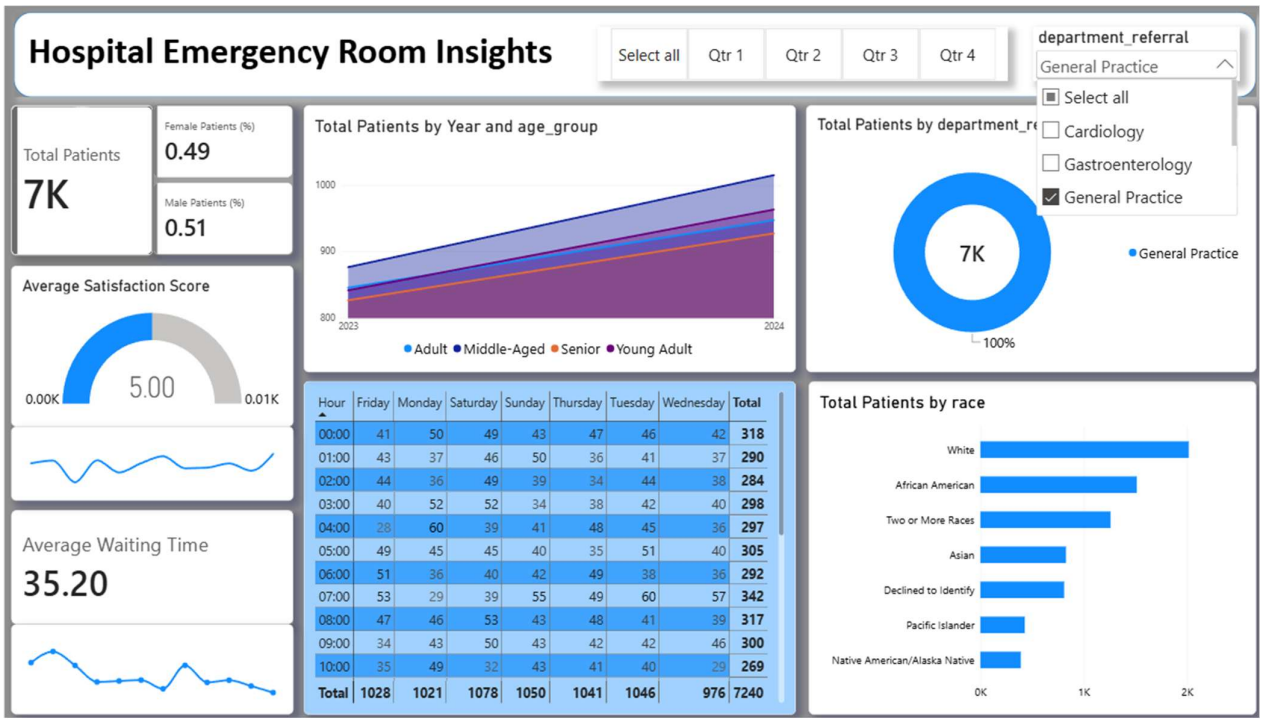
Finally, an interactive dashboard was developed in Power BI to present hospital emergency room insights visually.

The dashboard includes KPI cards for total patients, admitted patients, admission rate, average waiting time, and average satisfaction score. Charts display patient distribution by age group, gender, race, and department referral. Line and area charts help analyze patient trends over time, while a matrix heatmap highlights patient visit patterns by day and hour.

Slicers for department, gender, age group, admission status, and date allow users to filter the report and explore specific categories.

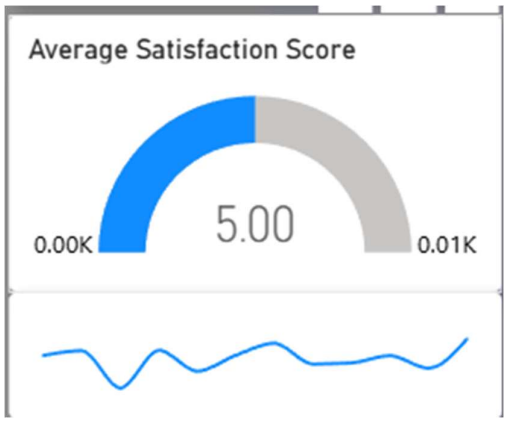


Hour	Friday	Monday	Saturday	Sunday	Thursday	Tuesday	Wednesday	Total
00:00	41	50	49	43	47	46	42	318
01:00	43	37	46	50	36	41	37	290
02:00	44	36	49	39	34	44	38	284
03:00	40	52	52	34	38	42	40	298
04:00	28	60	39	41	48	45	36	297
05:00	49	45	45	40	35	51	40	305
06:00	51	36	40	42	49	38	36	292
07:00	53	29	39	55	49	60	57	342
08:00	47	46	53	43	48	41	39	317
09:00	34	43	50	43	42	42	46	300
10:00	35	49	32	43	41	40	29	269
Total	**1028**	**1021**	**1078**	**1050**	**1041**	**1046**	**976**	**7240**



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6. Business Recommendations

- **Improve Patient Flow:** Monitor waiting times and identify periods when patient demand is high.
- **Plan Resources Effectively:** Use day-and-hour patient volume patterns to support staffing and resource planning.
- **Monitor Department Demand:** Review referral counts to understand the distribution of patient visits across departments.
- **Improve Data Quality:** Reduce missing department and satisfaction information to make reporting more complete.
- **Track Patient Experience:** Monitor satisfaction scores and waiting-time trends over time.
- **Use Interactive Reporting:** Apply Power BI filters to compare patient groups and investigate patterns in emergency room activity.